MIL-L-38200B(USAF)
22 September 1966
SUPERSEDING
MIL-L-38200A(USAF)
11 February 1966

MILITARY SPECIFICATION

LIFE PRESERVER, UNDERARM, ADULT LPU-5/P

1. SCOPE

1.1 This specification covers requirements for one type of inflatable life preserver that is designated LPU-5/P.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

V-T-295	Thread, Nylon
PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple-Wall
PPP-T-60	Tape; Pressure-Sensitive Adhesive, Waterproof for Packaging

Military

MIL-W-530	Webbing and Tape, Textile, Cotton, General Purpose, Natural or in Colors
MIL-B-543	Buckles, Tongueless and Web Strap
MIL-C-601	Carbon Dioxide, Technical, Cylinders of, for Inflation of Life Preservers
MIL-W-1053	Whistle Ball, Plastic
MIL-S-2785	Shark Repellant Compound, Packet
MIL-C-5024	Compound, Lubricating, Inner Tube, Aircraft Tire
MIL-C-5040	Cord, Nylon
MIL-A-5540	Adhesive, Polychloroprene
MIL-V-5774	Valve, Oral Inflation
MIL-T-8363	Tape and Webbing, Textile, Woven, Nylon
MIL-S-17980	Sea Marker, Fluorescein, Dye, Sodium Salt Type, Life Preserver Packet

FSC 4220

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MIL-C-19002	Cloth, Coated; and Tape, Coated Cloth-Chloroprene on Nylon, Pneumatic Life Preserver
MIL-C-19377	Cloth, Nylon, Twill, Inflatable Life Preserver
MIL-F-21840	Fastener Tape, Hook and Pile, Nylon
MIL-C-23070	Cloth, Laminated, and Tape, Coated Cloth, Natural Rubber on Nylon
MIL-I-23145	Inflation Assemblies, Life Preserver

STANDARDS

Federal

FED-STD-151	Metals: Test Methods
FED-STD-601	Rubber, Sampling and Testing
FED-STD-751	Stitches, Seams, and Stitchings

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MS20230	Grommets, Metallic, Plain and Spur, With Washers, Type I and Type III
MS33586	Metals, Definition of Dissimilar

Air Force-Navy Aeronautical

AN227	Fasteners - Snap and Curtain

DRAWINGS

Air Force

52B6586	Tube - Oral Inflation, Pneumatic Survival Equipment
52A6598	Stem - Inflator Valve, Life Preserver Bladder
52A6600	Cap - Inflator Valve, Pneumatic Life Preserver
53B6265	Knob - Pull Cable, Parachute Life Preserver
58A3568	Gasket - Life Preserver Inflator [illegible], Upper
58A3569	Gasket - Life Preserver Inflator [illegible], Lower

Bureau of Ships

S3306-1, 183395	Rubber Sleeve Check Valve for CO_2 Inflatable Life Preserver
608-1895222	Clamp, Brass

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Interstate Commerce Commission Regulation No. 13

(Copies may be obtained from T. C. George, Agent, 63 Vessey St., New York 7, New York.)

3. REQUIREMENTS

3.1 Qualification. The life preservers furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.4).

3.2 Components. The preserver shall consist of the following major components:

Item No.	Quantity	Item Name
1	2	Flotation cells
2	1	Container
3	1	Storage container
4	1	Emergency signal light
5	1	Whistle
6	1	Sea dye marker
7	1	Shark deterrent
8	1	Life line

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials

3.4.1 Elastomeric or rubber materials. Unless otherwise specifically approved by the procuring activity, all elastomeric or rubber materials and component parts used in the life preserver shall have been manufactured within 12 months of the date of delivery to the Government.

3.4.2 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to salt spray or atmospheric conditions likely to be met in storage or normal service.

3.4.2.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

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*3.4.3 Flotation cell cloth. The tape used in the construction of the cells shall conform to type II, III, or IV of MIL-C-19002. The coated cloth used in the construction of the flotation cells shall conform to type I of MIL-C-19002 with the exceptions specified in 3.4.3.1. All defects in the cloth coating not specifically covered by the applicable table of MIL-C-19002 shall be reinforced on the coated side with 1-inch-diameter discs of tape. Not more than four such discs shall be permitted in any cell. The disc shall be applied by cementing.

*3.4.3.1 Exceptions to MIL-C-19002. The following exceptions shall apply to the type I coated cloth covered by MIL-C-19002.

(a) The paragraph entitled Resistance to Accelerated Weathering in MIL-C-19002 shall be changed to specify the following: After exposure of the coated cloth to the 24 hour accelerated weathering test, by exposing the uncoated side of the cloth to the light source, the breaking strength shall be not less than 145 pounds in the warp direction and 130 pounds in the filling direction and the ultimate elongation shall be not less than 22 percent in the warp and the filling directions when tested by the ravel strip method. Specimens which are tested by the grab method shall have a minimum breaking strength of 80 pounds in the warp direction and 90 pounds in the filling direction and an ultimate elongation of 20 percent in either direction after weathering. A colorfastness to light equal to the standard color sample shall be maintained after weathering. (Note: Access to the standard color sample may be obtained as specified in 6.3.)

(b) The paragraph entitled Base Cloth Defects of MIL-C-19002 shall be changed to specify the following: After the coating process, the number of visible defects which are attributable to the base cloth shall not exceed the number existing in the base cloth before the coating process. Defects which are attributable to the base cloth shall not be considered as coated fabric defects. Critical coating defects shall not be accepted. All coating defects in an accepted roll of finished coated cloth shall be marked on the defect and at the selvage edge adjacent to the defect.

(c) In the paragraph entitled Base Cloth of MIL-C-19002, the sentence that states the base cloth shall conform to the requirements of Specification MIL-C-19377(Aer) shall be modified to read as follows: The base cloth shall conform to the requirements of MIL-C-19377 except that any amendments made to MIL-C-19377 shall not apply and that in lieu of the subparagraphs listed under the heading Acceptable Tests and the table entitled List of Defects, the following shall apply:

(1) The cloth shall be subjected to the quality conformance inspection. The quality conformance inspection shall consist of testing samples from each lot for conformance to section 3 of the specification and an examination of one sample of the filled shipping container from each lot for conformance to the packaging, packing, and marking requirements. A lot shall consist of cloth made essentially under the same conditions and in a single production run. In addition, a sample shall be selected from each lot in accordance with MIL-STD-105 and shall be run over a bank of lights having sufficient intensity to reveal defects in the cloth at a distance of 2 feet. All visible irregularities, including burl marks, shall be considered defects and shall be graded as specified herein.

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(2) Defects up to 1/2 inch in any dimension shall be graded one point. Defects more than 1/2 inch but not more than 3 inches in any dimension shall be graded two points. Defects more than 3 inches but not more than 6 inches in any dimension shall be graded three points. Defects exceeding 6 inches to full width or one running yard shall be graded four points. For the purpose of evaluation under MIL-STD-105, the number of points shall be defined as the number of defects. The number of points or defects shall be evaluated using an acceptable quality level of 150 defects per 100 yards.

(3) Three-point and four-point defects shall be flagged in the selvage for removal in the fabrication process. All imperfections protruding from the surface of the cloth such as knots, fuzz balls, slubs, and kinky filling shall be removed by careful burling before the cloth is offered for delivery.

3.4.4 Container cloth. The coated cloth used in the construction of the container housing the cells and the storage container shall be the same as that used for the construction of the cell.

3.4.5 Reinforcing cloth. All reinforcing cloth shall conform to variety C of MIL-C-23070.

*3.4.6 Fastener tape. The fastener tape for the container shall be either 1/2 inch or 5/8 inch wide and shall conform to type I of MIL-F-21840, except that the color shall be as specified in 3.9.

3.4.7 Chafing pad. The coated cloth used in the construction of the chafing pad shall conform to variety C of MIL-C-23070.

3.4.8 Harness webbing. The harness webbing including the chest strap shall be 3/4-inch-wide tape conforming to type IIa of MIL-W-530.

3.4.9 Thread. Unless otherwise specified, the thread used in the construction of the preserver shall conform to type I or II, class 1, size E of V-T-295.

3.4.10 Cotton materials. Cotton materials used in the construction of the preserver shall be treated with a mildew inhibitor that is not deleterious to any material used in the preserver.

*3.4.11 Cement. The cement used in the construction of the flotation cells shall conform to class 1, 2, or 3 of MIL-A-5540.

3.5 Design and construction. Two superimposed flotation cells shall be permanently attached to a container and to each other and an adjustable harness. An adjustable strap with a snap and ring connection shall be provided to connect the ends of the cells. The two cells, an emergency signal light, a whistle, a sea dye marker, a shark deterrent, and a life line shall be assembled and placed in a storage container, as specified in 3.5.1.

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3.5.1 Unit assembly. The preserver shall be assembled as follows: The flotation cells shall be completely deflated by means of a vacuum source through the oral tube and valve assembly. The inflator assemblies with gaskets, caps, chaffing pad, and charged carbon-dioxide cylinders shall be mounted on the inflator valve stem of each flotation cell. The carbon-dioxide cylinders shall be inserted into the cylinder holders on each cell. The emergency signal light shall be attached by snap fasteners to the right side of the upper cell. The shark deterrent shall be tied to the cylinder holder on the lower cell, and the sea dye marker shall be tied to the cylinder holder on the upper cell. The whistle shall be tied to the cell-connecting-strap hardware on either cell, and the lanyard shall be faked and secured by a rubber band to prevent entanglement during inflation. The life line shall be tied to the "D" ring on the flotation cells, and the remainder shall be faked and secured by rubber bands to prevent entanglement during inflation. The components shall be positioned in the container in a way to provide an even closure of the container. The flotation cells shall be folded to position the inflators near the ends of the container and to allow ease of operation of the inflator. The inflator lanyards shall be routed through the lower corners of the container closure, and the closure pins shall be routed through the sleeves of the end flaps. The fastener tape shall be connected at the approximate center of the container, and the closure shall be continued towards the ends of the case. The flaps shall be folded over the ends of the container, and the nylon loops shall be pulled through the grommets in the flaps. The closure pins shall be inserted in the loops and shall be safetied with a single thread conforming to type I or II, class 1, size B of V-T-295. The completely assembled preserver shall be folded and placed in the storage container.

NOTE: All preservers shall have charged carbon-dioxide cylinders installed prior to final assembly (see 3.6.4).

3.5.2 Operation of preserver. When the complete preserver is properly assembled and donned, a pull on the knobs of the inflation lanyard shall operate the preserver as follows:

(a) Break the safety ties on the closure pins

(b) Withdraw the closure pins from the loops

(c) Actuate the inflator levers to inflate the flotation cells which separate the tape around the container

(d) The shark deterrent, the sea dye marker, and th signal light battery shall drop below the lower cell and, when in actual service, shall enter the water.

3.5.3 Cementing. Either cementing or pressure-curing by heat shall be used in the construction of seams, sealing areas, and attachments of all component parts of the preserver. All surfaces to be cemented shall be clean. All cement used shall be fresh and shall be mixed within 8 hours of application. Fresh cement shall not be put into containers containing congealed cement. The cement shall be applied over the total contact areas and shall extend beyond all seam tapes and attachments by at least

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1/8 inch on the inside and the outside of the cells and not more than 3/8 inch on the outside of the cells. After cemented areas are joined, trapped air and wrinkles shall be rolled out. The interior of each cell shall be dusted lightly with compound conforming to MIL-C-5024 to prevent adherence of noncemented surfaces.

3.5.4 Stitching. Unless otherwise indicated, the stitches used in the construction of the preserver shall conform to stitch type 301 of FED-STD-751, with 8 to 10 stitches per inch. All seams shall be backstitched at least 1/2 inch. Stitching shall not be used in the gas-retaining portion of the flotation cells.

3.6 Performance

* 3.6.1 Operation. After the life preserver has been properly assembled as specified in 3.5.1, it shall be donned by an individual who shall simultaneously actuate the respective cells. The carbon-dioxide cylinders shall be discharged when a reasonable pull is applied to the respective lanyards. The respective flotation cells shall inflate, in the time specified, to the following minimum pressures: 0.15 pounds per square inch (psi) 20 seconds after discharge of the cylinder and 1.0 psi 45 seconds after discharge of the cylinders. The maximum pressure in the cells shall not exceed 3.0 psi when the temperature of the flotation cells reaches plus 70 (plus or minus 2) degrees Fahrenheit. The cells shall not leak after this test.

3.6.2 Pressure. Each flotation cell shall withstand an inflation pressure of 5 psi for a period of 5 minutes without evidence of failure of the materials or cemented areas or without evidence of permanent distortion of the cell from its designed shape.

* 3.6.3 Leakage. Each flotation cell shall be inflated to 2 psi. After 24 hours the pressure in the cells shall not be less than 1-1/2 psi.

3.6.4 Cylinder weight. Before the cylinder is installed on the preserver, the cylinder and the carbon-dioxide charge shall not weigh less than the gross weight marked on the cylinder.

3.6.5 Burst. The flotation cell shall not burst when subjected to an inflation pressure of 12 psi.

3.6.6 Oral valve attachment. The oral valve shall be attached to the oral tube to withstand a load of 50 pounds.

3.6.7 Color matching. When examined visually, the color of the materials used in the preserver shall approximately match the color of the standard color samples (see 3.9 and 6.3).

3.6.8 Adhesion. The adhesion of all seams and attachments to the flotation cells shall be not less than 5 pounds per inch of width after joining coated to uncoated sides of the cell cloth and not less than 7-1/2 pounds per inch of width after joining coated to coated sides of the cell cloth.

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3.6.9 Leakage after operation. After the preserver has been inflated ten times as specified in 4.7.10, the cells shall not leak.

3.6.10 Buoyancy. When the two flotation cells of a completely assembled preserver are inflated by the carbon-dioxide cylinders, the preserver shall support a weight of 50 pounds in water.

3.6.11 Salt spray. Unless otherwise specified, all metal parts used in the construction of the preserver shall withstand a 96-hour exposure to salt spray without corrosion of the base metal.

3.7 Details of components

* 3.7.1 Flotation cells. The shape and dimensions of the flotation cells shall conform to figure 1. Each cell shall consist of not more than two panels joined around the perimeter to meet the requirements specified herein. Each cell shall have a mechanical inflation system and an oral inflation system. Internal patches of cell cloth shall be used to reinforce the attachment of the inflation systems. Holders made of harness webbing shall be cemented to the cells to retain the carbon-dioxide cylinders when they are inserted in the mechanical inflators. A snap fastener shall be cemented to the right side of the upper cell for attachment of the signal light. Each cell shall have an internal fabric baffle to prevent cold crocking of the cell fabric when the carbon-dioxide gas enters.

* 3.7.1.1 Mechanical inflation system. The mechanical inflation system shall consist of the following:

(a) Lanyard assembly in accordance with figure 2

(b) Two carbon-dioxide cylinders conforming to type II of MIL-C-601

(c) Inflator stem conforming to Drawing 52A6598-1 with a core conforming to Drawing S3306-1, 183395

(d) Valve stem cap conforming to Drawing 52A6600

(e) Upper gasket conforming to Drawing 58A3568

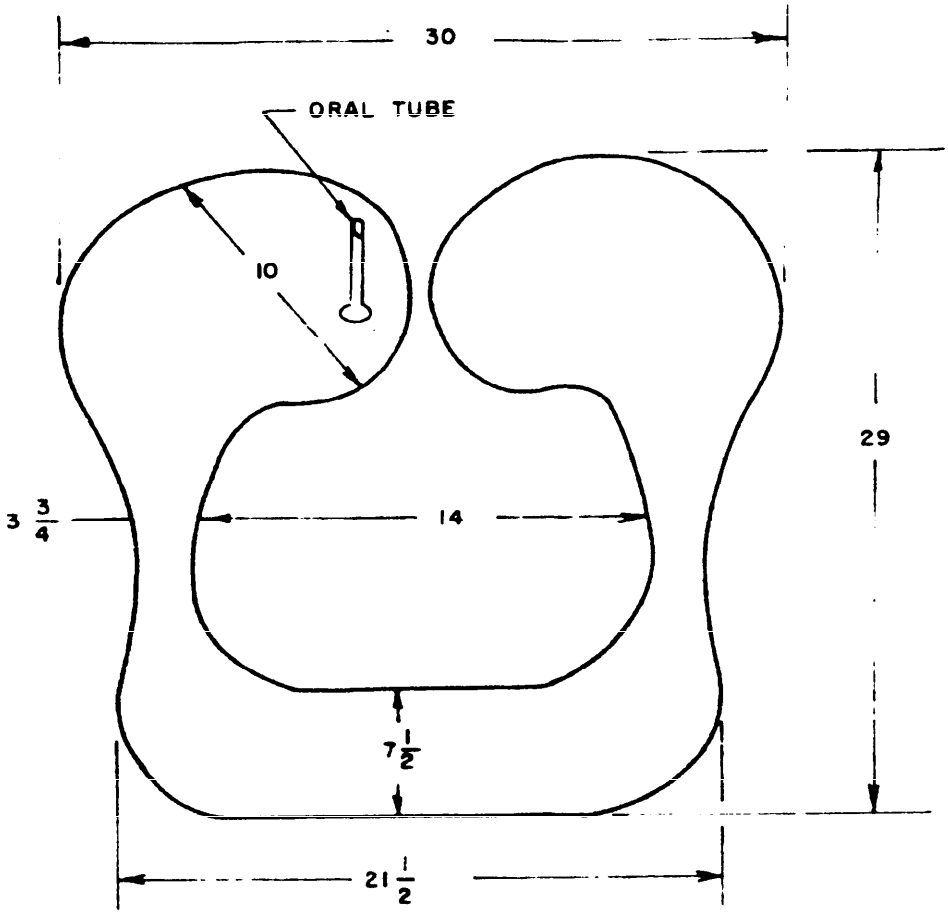
(f) Lower gasket conforming to Drawing 58A3566

(g) A chaffing pad conforming to figure 3

(h) An inflator conforming to type III of MIL-I-23145 with the following exceptions:

(1) In the paragraph entitled Piercing Pin of MIL-I-23145, the requirement that the normal position of the point of the piercing pin shall be a minimum of 0.010 inch

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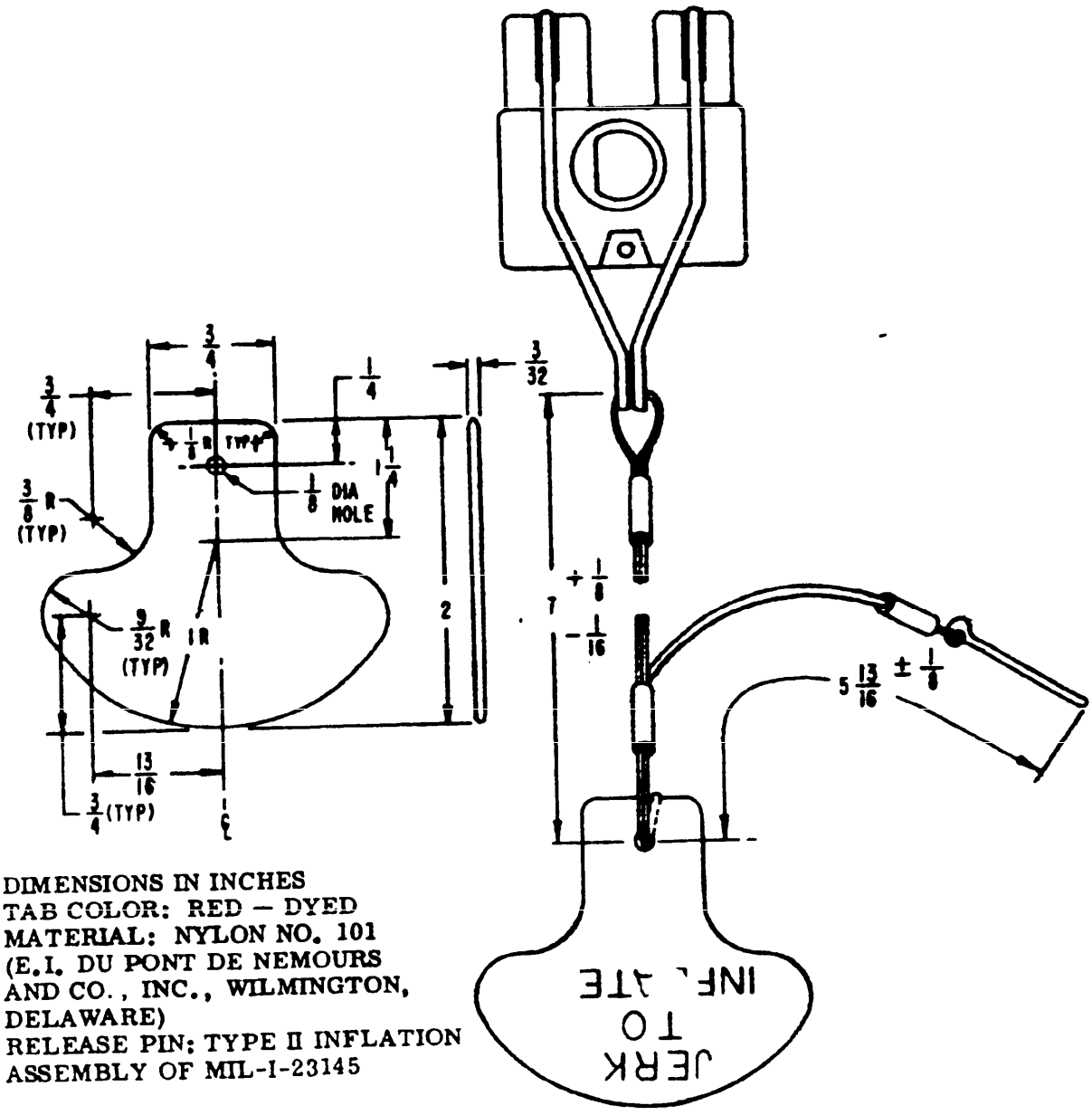


DIMENSIONS IN INCHES

DIMENSIONS ARE APPROXIMATE AND ARE INTENDED ONLY AS A GUIDE TO THE DESIRED SHAPE
PRESSURE VERSUS TIME REQUIREMENTS OF 3.6.1 SHALL DETERMINE THE ACTUAL VOLUME AND THE DIMENSIONS OF THE CELL.

* Figure 1. Flotation Cell

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DIMENSIONS IN INCHES
TAB COLOR: RED — DYED
MATERIAL: NYLON NO. 101 (E.I. DU PONT DE NEMOURS AND CO., INC., WILMINGTON, DELAWARE)
RELEASE PIN: TYPE II INFLATION ASSEMBLY OF MIL-I-23145

KNOB CONFORMING TO 53B6265 MAY BE USED IN LIEU OF PULL TAB
MEASURE DIMENSIONS FOR LANYARD LENGTH FROM TOP OF KNOB

Figure 2. Lanyard Assembly

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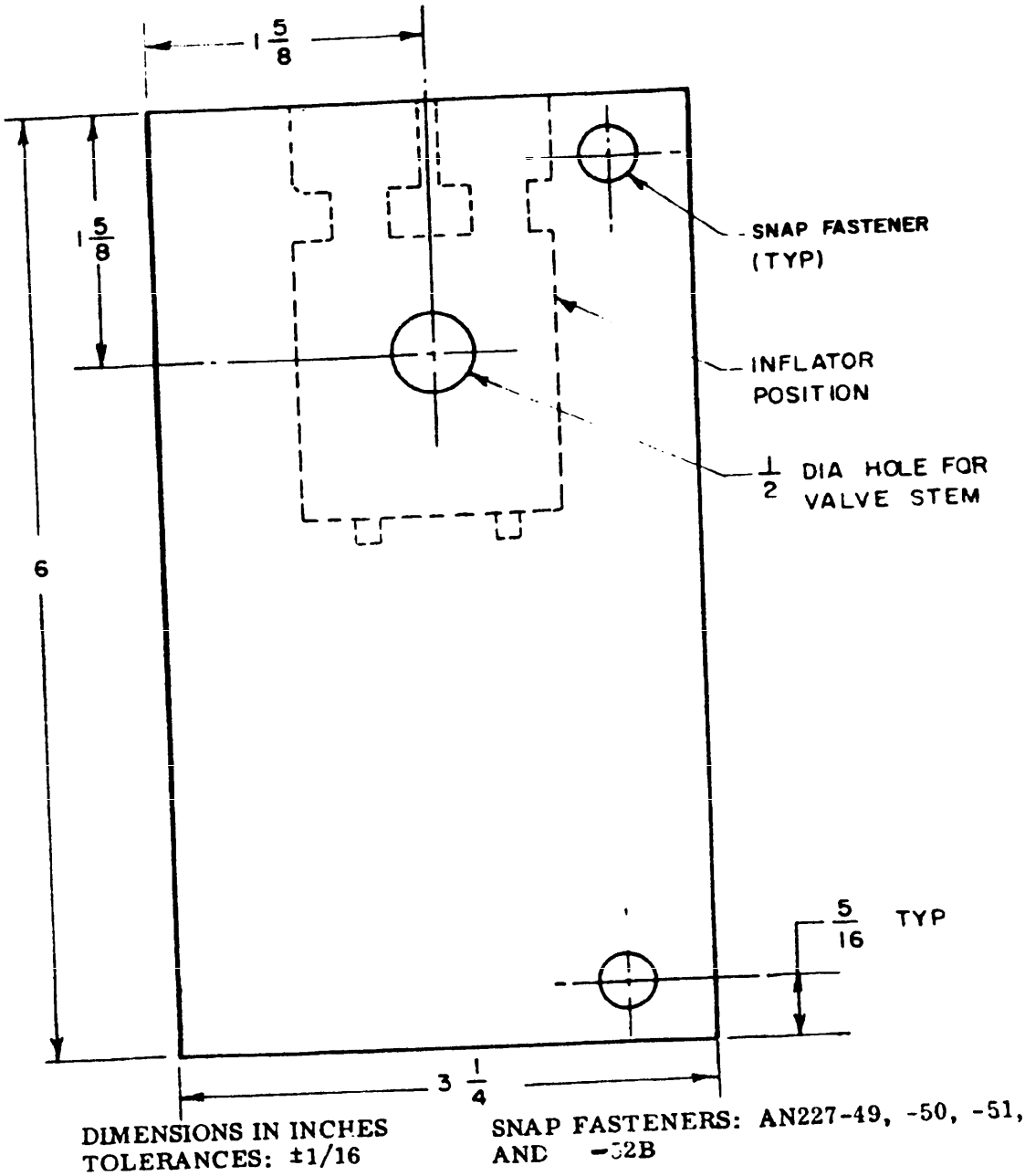


Figure 3. Chafing Pad

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below the upper surface of the seat seal gasket shall not apply. In lieu of this requirement, the normal position of the upper surface shall be a minimum of 0.005 inch below the upper surface of the metal seat of the casting.

(2) In the paragraph entitled Piercing Pin of MIL-I-23145, the requirement that the point of the piercing pin shall travel 0.090 to 0.120 inch beyond the upper surface of the seat seal gasket shall not apply. In lieu of this requirement the point of the piercing pin shall travel 0.100 to 0.130 inch beyond the upper surface of the metal seat of the casting upon actuation.

(3) In lieu of the paragraph entitled Piercing Pin Position and Travel Lever Operation in section 3 of MIL-I-23145, the following shall apply: When tested as specified herein, the point of the piercing pin shall be a minimum of 0.005 inch below the upper surface of the metal seat of the casting and shall advance 0.100 to 0.130 inch beyond the upper surface of the metal seat and shall retract to its original position with a travel of the lever through an arc not greater than 130 degrees in line with the longitudinal axis of the body.

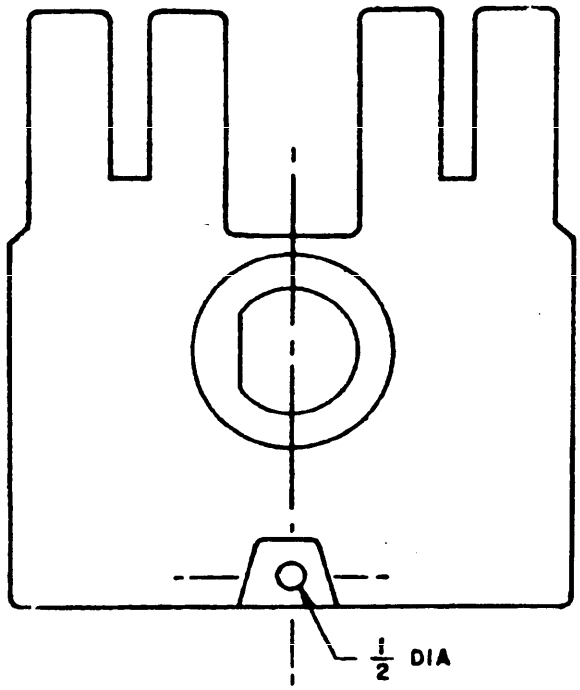
(4) In lieu of the paragraph entitled Piercing Pin Position and Travel and Lever Operation in section 4 of MIL-I-23145, the following shall apply: An accurately calibrated gage shall be used to determine that the point of the piercing pin in its normal position is a minimum of 0.005 inch below the upper surface of the metal seat of the casting and advances of 0.100 to 0.130 inch beyond the upper surface of the metal seat and retracts to its original position with a travel of the lever through an arc not greater than 130 degrees in line with the longitudinal axis of the body. The seat seal gasket can be removed for this test.

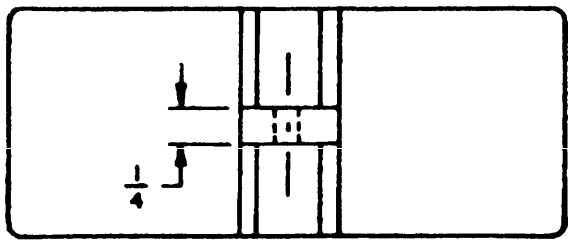
(5) The figure entitled Type III Inflation Assembly in MIL-I-23145 shall be changed to provide for safety wire attachment in accordance with figure 4 of this specification.

(6) The figure entitled Sections A-A and B-B, Set Screw and Seat Seal for Type III Inflation Assembly in MIL-I-23145 shall be changed as follows: Include provisions for safety wire attachment in accordance with figure 4 of this specification. Change the distance between piercing pin centers from 7/8 inch to 15/16 inch. Change the maximum width of the casting from 1-1/2 inches to 1-9/16 inches. Change the thickness of the seat seal gasket from 0.095 inch to 0.030 inch. C[illegible]nge the thickness of the seat seal gasket undercut from the present size to a size [illegible]t will accommodate a 0.30-inch gasket.

3.7.1.2 Oral inflation system. The oral inflation system shall consist of an oral tube conforming to Drawing 52B6586-1-3-1/4 and an oral valve conforming to type I of MIL-V-5774. The valve shall be inserted in the tube and attached with a clamp conforming to Drawing 608-1895222. The tube shall be positioned so that it points away from the user when the preserver is inflated.

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DIMENSIONS IN INCHES

Figure 4. Modification of Inflator (Attachment of Safety Wire)

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* 3.7.2 Container. The container shall conform to figure 5: The body of the container shall be constructed of one panel, folded in the middle to form an envelope and held closed with fastener tape. The end flap shall have a sleeve to cover the closure pin and inflation lanyard. A grommet conforming to MS20230B10 shall be installed in each flap and positioned so that a smooth closure of the container is effected when the preserver is assembled as specified in 3.5.1. The installation of the grommets shall be reinforced with the cloth specified in 3.4.5. The reinforcement cloth shall be boxstitched to the container. A loop of tape to hold the flaps closed when the closure pin is inserted shall be boxstitched to the side of the container. The tape utilized for the closure loops shall conform to type I of MIL-T-8363.

3.7.2.1 Harness. An adjustable harness shall be stitched to the container as shown on figure 5. All points of attachment shall be reinforced with the cloth specified in 3.4.5, and this reinforcing cloth shall be boxstitched to the container. The harness shall consist of two shoulder and back straps joined together at the rear of the container with a double thickness Y-shaped or V-shaped piece of container cloth and an adjustable chest strap attached to the inside of the container. The adjustability of the shoulder straps shall be 6 to 26 inches. The adjustability of the chest strap shall be 24 to 60 inches. The straps of the harness shall be marked for identification. The adjustment fittings shall have no sharp edges or burrs and shall be tongueless, 2-bar, buckles conforming to type II, style 4, class 4, size 3/4 inch of MIL-B-543.

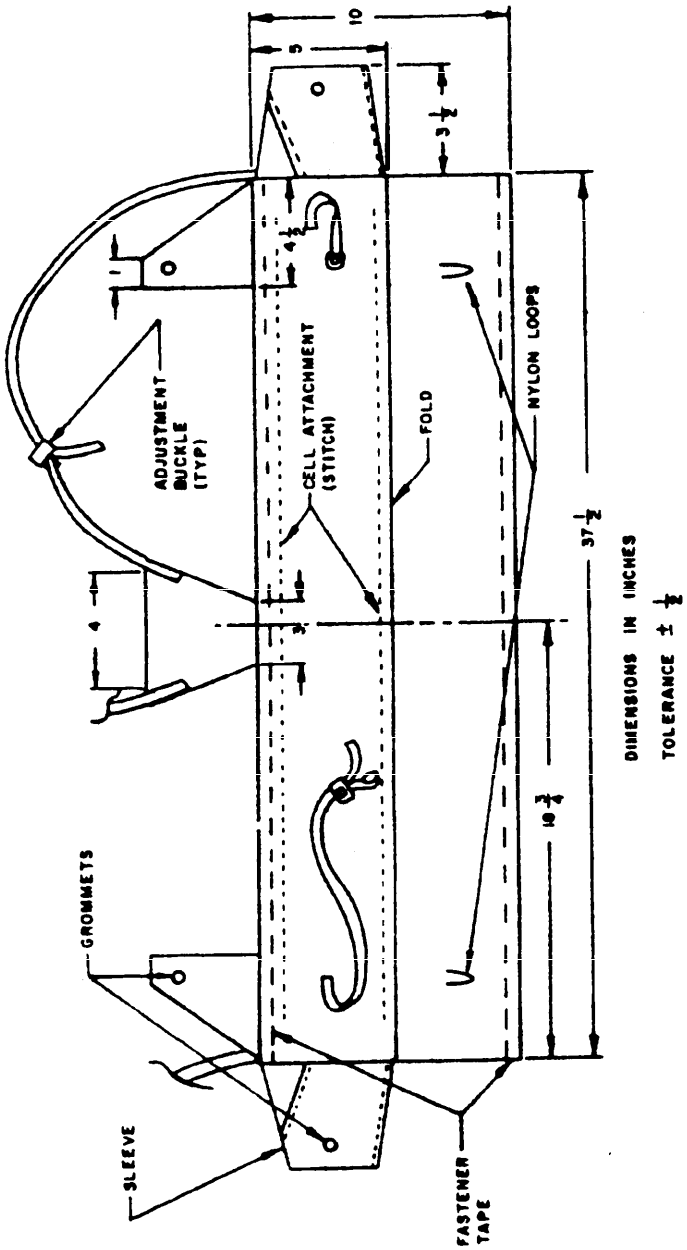
3.7.2.2 Container to cell attachment. The flotation cells shall be permanently attached to the container by stitching along the entire inner side of the container as shown in figure 5. The attachment shall be reinforced by the cloth specified in 3.4.5 at least 2 inches from each end of the stitching. The coated side of the container cloth shall be toward the flotation cells.

3.7.3 Storage container. The complete preserver, assembled as specified in 3.5.1, shall be packaged in a storage container conforming to figure 6. A pocket shall be sewn on the rear exterior of the storage container. The pocket shall accommodate an inspection card measuring 3-1/4 inches by 3-1/2 inches by 1/32 inch thick. The card shall not be furnished with the preserver.

3.7.4 Emergency signal light. The emergency signal light shall consist of a signal bulb, a water-activated battery, and electric cord, and an attachment patch and shall be attached to the snap-fastener assembly on the flotation cell. The emergency signal light used shall be one that has been approved by the procuring activity (see 6.5).

* 3.7.5 Whistle. The whistle shall conform to type II, either split or solid construction of MIL-W-1053.

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NOTE: THE ATTACHMENT POINTS FOR THE SHOULDER AND THE CHEST STRAPS SHALL BE DETERMINED BY THE ADJUSTABILITY REQUIREMENTS OF 3.7.2.1.

Figure 5. Container and Harness

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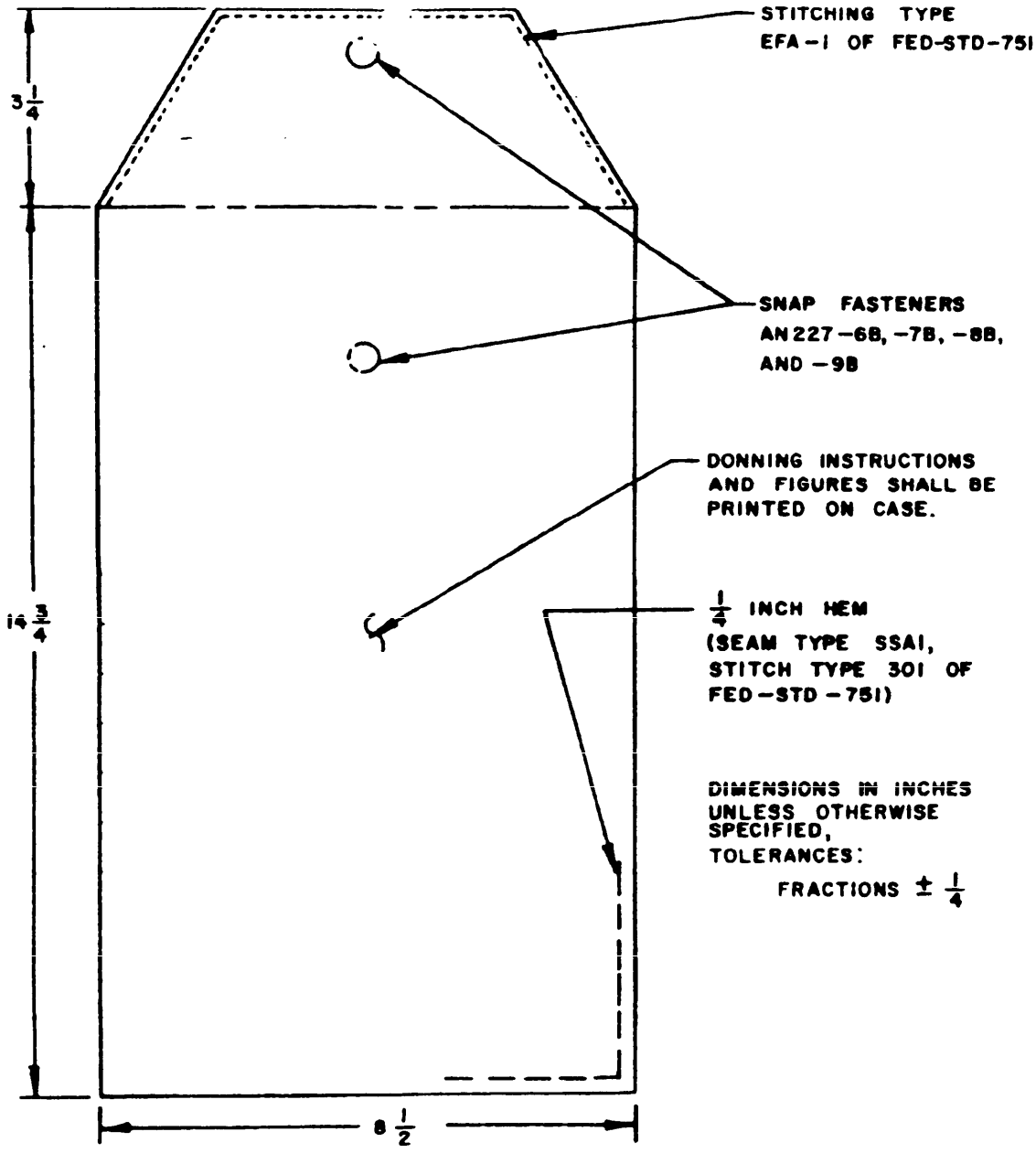


Figure 6. Storage Container

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3.7.6 Sea dye marker. The sea dye marker shall conform to MIL-S-17980, except that the attaching tape shall be replaced with a 12-inch length of cord conforming to type I of MIL-C-5040. The ends of the cord shall be seared to prevent fraying.

3.7.7 Shark deterrent. The shark deterrent shall conform to MIL-S-2785, except that the attaching tape shall be replaced with a 12-inch length of cord conforming to type I of MIL-C-5040. The ends of the cord shall be seared to prevent fraying.

* 3.7.8 Life line. The life line to attach to other survivors shall be a 10-foot length of cord conforming to type I of MIL-C-5040. A 1/4-inch swivel eye snap equal to Part No. 344 snap (manufactured by the North & Judd Manufacturing Co., New Britain, Connecticut) and free from burrs and sharp edges shall be securely attached to one end. The ends of the cord shall be seared to prevent fraying.

3.8 Weight. Excluding the emergency signal light, the whistle, the sea dye marker, the shark deterrent, and the life line, the complete preserver shall not weigh more than 2.5 pounds. The cells, excluding the mechanical inflator and the cylinders, shall not weigh more than 0.60 pound each.

3.9 Color. The color of the cells and container shall be molten orange. Unless otherwise specified (see 6.7) the color of all webbing, cord, and thread shall closely match the color of the cells and the container. The color of the exposed cement shall blend with the color of the flotation cloth. The color of the fastener tape shall be Shade No. 770 (Velcro Sales Corp., 681 Fifth Ave., New York, New York).

3.10 Identification of product. Unless otherwise specified, all markings shall be made with a durable nonmigrating ink of a color to contrast with the color of the surface marked. The ink shall have no deleterious effect on the surface marked. The marking shall conform to MIL-STD-130.

* 3.10.1 Preserver. The preserver shall be marked in 1/4-inch letters and numerals on the outside rear of container as follows:

Life Preserver, Underarm, Adult, LPU-5/P

Spec MIL-L-38200B(USAF)

Manufacturer's Identification_____

Contract or Order No._____

Date of Manufacture_____

Serial No._____

3.10.2 Upper flotation cell. The following information shall be marked in 1/4-inch letters, at the location of the applicable component on the upper flotation cell. One-inch-long arrows shall point to the components.

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(a) Signal light - To activate: Pull plug(s) from battery

(b) Sea dye marker - Pull tab to release

(c) Shark chaser - Pull tab to release

(d) Whistle

(e) Cell connector - Connect snap to ring and pull to adjust

(f) Life line - Attach to other survivor life preserver

(g) Oral valve - To increase cell pressure unscrew knurled knob (counterclockwise) and depress. Immediately blow through tube.

* 3.10.3 Storage container. The storage container shall be marked with vivid illustrations detailing the donning instructions and depicting the correct appearance of the preserver and the location of the components when donned and operated properly. The top flap of the storage container shall be marked in 1/2-inch letters with the word ADULT. The lower front of the storage container shall be marked in 1/4-inch letters and numerals with the following information:

Life Preserver, Assembly

Type LPU-5/P

Spec MIL-L-38200B(USAF)

Manufacturer's P/N_____

Manufacturer's Name_____

Serial No._____

3.11 Workmanship. Workmanship shall be in accordance with high grade manufacturing practices covering this class of work. Because of the emergency use of this equipment, all details of workmanship shall be of the highest quality to insure proper functioning of the preserver at all times.

4. QUALITY ASSURANCE PROVISION

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of tests. The inspection and testing of life preservers shall be classified as follows:

(a) Qualification tests

(b) Acceptance tests

(c) First article inspection.

4.3 Test conditions

* 4.3.1 Atmospheric conditions. Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of 23 to 32 inches of mercury, at a temperature of 70 (plus or minus 18) degrees Fahrenheit, and at a relative humidity of 80 percent or less. Where tests are concluded with atmospheric pressure or temperature differing from the starting values, corrections shall be made for the change in values.

* 4.3.2 Temperature corrections. If the ambient temperature changes during the 24-hour leakage test, adjustment shall be made on the basis of a change in pressure of 0.0312 psi per degree Fahrenheit change in temperature.

4.3.3 Barometric corrections. If the barometric pressure changes during the leakage test, adjustment shall be made on the basis of change in pressure of 0.489 psi per inch (mercury) change in barometric pressure.

4.4 Qualification tests. The qualification test samples (life preservers and materials) shall be subjected to the qualification tests specified in 4.4.2.

4.4.1 Test samples. The qualification test samples shall consist of two life preservers, material samples, and reports as specified in the letter of authorization.

4.4.2 Tests. Qualification tests shall consist of all tests described under 4.7.

4.5 Acceptance tests. Acceptance tests shall consist of:

(a) Individual tests

(b) Sampling tests.

4.5.1 Individual tests. Each life preserver shall be subjected to the following tests as described under 4.7:

(a) Examination of product

(b) Operation

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(c) Pressure

(d) Leakage

(e) Cylinder weight.

* 4.5.1.1 Rejection. The preservers may be corrected and submitted to the procuring activity for acceptance, but an air-retaining seam shall not be reworked, under any circumstances, more than one time.

4.5.2 Sampling plans and tests

4.5.2.1 Lot. A lot shall consist of life preservers manufactured under essentially the same conditions and submitted for inspection at substantially the same time. A lot shall not exceed 200 life preservers for sampling plans A and B and shall be in accordance with inspection level S-3 of MIL-STD-105 for sampling plan C.

4.5.2.2 Sampling plan A. One life preserver selected at random from each lot of 200, or fraction thereof, shall be subjected to the following tests as described under 4.7:

(a) Burst

(b) Oral valve attachment

(c) Color matching.

4.5.2.2.1 Rejection, plan A. When one or more items from a lot fail the specified tests, this shall be cause for rejection of the entire lot. Before resubmitting the lot, the contractor shall fully explain to the inspector both the corrections made and the cause of previous rejections. Preservers that have been rejected may be replaced or repaired and resubmitted for acceptance. Preservers rejected after resubmission shall not again be resubmitted without specific approval of the procuring activity. An air-retaining seam shall not be reworked after the preserver is submitted for acceptance.

4.5.2.3 Sampling plan B. An adhesion test sample shall be prepared from cell fabric by each worker each time a different batch of cement is used by the worker. The sample shall be identified with the lot in production. Five adhesion test samples shall also be prepared from each incoming roll of cell fabric. The cementing procedure shall be the same as that used in the construction of the flotation cells. In addition, five sample seams and five sample patches shall be cut from the flotation cell used in sampling plan A. These samples shall be subjected to the adhesion test specified in 4.7.9.

4.5.2.3.1 Rejection, plan B. When one or more of the adhesion test samples specified in 4.5.2.3 fail the specified test, production shall be halted until the cause of

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failure is determined and the corrections made. When a sample seam from a flotation cell fails the specified test, the lot represented shall be rejected. Lots rejected on the basis of seam-adhesion tests may be resubmitted for acceptance provided five additional flotation cells picked at random from the lot are tested for adhesion, and each is found to conform to requirements.

* 4.5.2.4 Sampling plan C. Upon completion of the individual tests specified in 4.5.1, random samples shall be selected from each lot in accordance with inspection level S-3 of MIL-STD-105. The sample size shall be based only on the applicable size code letter corresponding to the inspection level S-3. Each life preserver selected as a sample unit shall be forwarded to an inspection activity selected by the procuring activity (see 6.2 and 6.4) for the following tests and examination:

(a) Operation

(b) Pressure

(c) Leakage

(d) Examination of product.

4.5.2.4.1 Rejection, plan C. The acceptance number shall be zero, and the rejection number shall be one for each test and examination. The lot, from which the life preserver samples were selected, shall not be shipped until the samples have been approved by the procuring activity and the lot has been accepted by the cognizant quality control representative. The lot from which the life preserver sample was selected shall be rejected if any sample fails to meet any requirement of the tests specified in 4.5.2.4. A rejected lot shall not be resubmitted except with the approval of the procuring activity. Upon completion of the tests and examinations, the sample will be returned to the manufacturer.

4.6 First article inspection. Unless otherwise specified (see 6.2), the samples specified in 4.6.1 shall be submitted as specified in 4.6.1.1 for the performance of examinations and tests deemed necessary by the testing activity.

4.6.1 Samples. First articles samples shall consist of three complete life preservers of the type specified in the contract or order. The samples shall be representative of the construction, components, materials, and workmanship that will be used during production.

4.6.1.1 Submitting samples. Unless otherwise specified, as soon as practicable after the award of the contract or order, samples specified in 4.6.1 shall be forwarded as specified in the contract or order. When a contractor is in continuous production of life preservers from contract to contract, submission of further samples on the new contract may be waived at the discretion of the procuring activity. Approval of the first article samples or the waiving of the first article inspection does not preclude the requirements for performing the acceptance tests.

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*4.6.1.2 Identification of sample. The first article samples submitted shall be plainly identified by securely attached tags marked with the following information:

Sample submitted by (name) (date) for first article inspection in accordance with MIL-L-38200B(USAF) under Contract No._____.

4.6.1.3 Disposition of samples. Disposition of the first article samples shall be as specified by the procuring activity (see 6.2).

4.6.1.4 Results of inspections. Results of inspections, and where applicable, recommendations, and comments pertinent for use in monitoring production shall be kept available for the procuring activity.

4.7 Test methods

4.7.1 Examination of product. Each life preserver shall be visually examined to determine compliance with the requirements of this specification with respect to workmanship, materials, marking, and construction.

*4.7.2 Operation. The life preserver, assembled as specified in 3.5.1 shall be adjusted to an individual to simulate actual service use. The carbon-dioxide cylinders shall be discharged simultaneously with a reasonable pull to determine if the cells will come out of the container (see 3.5.2). The components shall not be tangled or in any way prohibit the cells from inflating to their designed shape. The minimum internal cell pressure 20 seconds after discharge of the cylinders shall be 0.15 psi. The minimum internal cell pressure 45 seconds after discharge of the cylinders shall be 1.0 psi. The maximum internal cell pressure shall not exceed 3.0 psi when the cell temperature has reached 70 (plus or minus 2) degrees Fahrenheit. Failure of the cells to inflate as specified or evidence of leakage or failure of the mechanical inflation assembly shall be cause for rejection. After this test is completed, the flotation cells shall be completely deflated by means of a vacuum source, and the life preservers shall be subjected to the test specified in 4.7.3.

4.7.3 Pressure. Each cell of the preserver shall be inflated to 5 psi with oil-free air. After a minimum of 10 minutes, the pressure shall be checked and readjusted, if necessary, to the original pressure of 5 psi. The temperature and pressure shall be recorded at this time. A minimum of 5 minutes after the readjustment period, the pressure shall be measured and corrected for any change in temperature or barometric pressure. The corrected pressure shall not be less than 4.60 psi. After these measurements and corrections, the pressure shall be released. Any evidence of failure of the material of the cemented areas or any evidence of permanent distortion of the cell shall be cause for rejection.

*4.7.4 Leakage. For this test, the carbon-dioxide cylinders shall not be attached. Each cell of the preserver shall be inflated with oil-free air to a pressure of 2 psi. After a minimum of 15 minutes, the pressure shall be checked and readjusted, if

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necessary, to the original pressure of 2 psi. The pressure source shall then be removed. After a minimum of 24 hours after the readjustment period, the pressure shall be measured and corrected for any change in temperature or barometric pressure. The corrected pressure shall not be less than 1-1/2 psi. There shall be no leakage from the valve stem. This test shall follow the test specified in 4.7.3. Failure to meet these requirements shall be cause for rejection.

4.7.5 Cylinder weight. Each cylinder shall be weighed not more than 72 hours prior to its installation on the preserver. The weight of the cylinders and carbon-dioxide charge shall not be less than the gross weight marked on the cylinder.

4.7.6 Burst. The flotation cells shall be subjected to an inflation pressure sufficient to cause failure of material or seam areas. The pressure at which failure occurs and the type of failure shall be recorded. Failure at a pressure of 12 psi or less shall be cause for rejection.

4.7.7 Oral valve attachment. The strength of attachment of the oral valve in the tube shall be determined by applying force along the longitudinal axis of the assembly in a tensile machine with a jaw separation of 12 inches (plus or minus 1/2 inch) per minute. Failure to withstand a load of 50 pounds shall be cause for rejection.

4.7.8 Color matching. The color of the cells and the container shall be compared to, and shall approximately match, the color of the standard shade (see 6.3). The color of the webbing, the thread, and the cords shall closely match the color of the cells and the fabric. Variations in the shade of individual materials that are apparent shall not be used together in the same unit.

4.7.9 Adhesion. Adhesion shall be determined in accordance with method 8011 of FED-STD-601.

4.7.10 Leakage after operation. The life preserver shall be subjected to ten operations as specified in 4.7.2. At the conclusion of these operations, the internal cell pressure shall be adjusted to 2-1/2 psi and the cells submerged in water for not less than 2 minutes. Any evidence of leakage of the cells or inflation assemblies shall be cause for rejection. Any nonconformance to the pressure requirements of 4.7.2 during the ten operations shall be cause for rejection.

4.7.11 Buoyancy. A completely assembled life preserver shall be inflated with carbon-dioxide cylinders. The discharged cylinders shall be left in the inflators. Each side of the flotation cells shall have a 25-pound steel weight tied by a 2-foot-length cord so that the 50 pounds is distributed evenly. The weighted preserver shall be placed in water and shall not submerge. Failure of the preserver to support the weights as specified shall be cause for rejection.

4.7.12 Salt spray. Each metal part or assembly not protected against corrosion by salt spray in accordance with Government specifications, standards, or drawings shall be subjected to a 96-hour salt spray test in accordance with method 811 of

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FED-STD-151. After this test the metal part or assembly shall be examined for corroded areas. A corroded area, which is defined as exposure or corrosion of the base metal, shall be cause for rejection of the part or assembly.

5. PREPARATION FOR DELIVERY

* 5.1 Preservation and packaging. Preservation and packaging shall be A or C as specified (see 6.2).

* 5.1.1 Level A. After the life preserver has been assembled and placed in the storage container as specified in section 3, it shall be individually packaged in a box conforming to class weather-resistant of PPP-B-636. All box joints and seams, including the manufacturer's joint, shall be sealed with waterproof pressure-sensitive tape conforming to class 3 of PPP-T-60. The tape shall be applied as specified in the appendix of PPP-B-636. Staples and stitches shall be covered on the inside surfaces of the box with tape conforming to class 1 of PPP-T-60 to protect the life preserver from abrasion.

* 5.1.2 Level C. Preservation and packaging shall be the minimum necessary to afford protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity.

*5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

* 5.2.1 Level A. Life preservers of like description shall be packed in boxes conforming to class 2, style optional of PPP-B-621 or overseas type, style A or B, class 2 for type 2 load of PPP-B-601. The boxes shall be strapped in accordance with the appendix of the applicable box specification. Selection of containers shall be at the option of the contractor. The gross weight of each container shall not exceed 200 pounds.

* 5.2.2 Level B. Life preservers of like description shall be packed in containers conforming to weather resistant, style optional of PPP-B-591; weather resistant, style optional of PPP-B-636; or class 2, grade A, style optional of PPP-B-640. The gross weight of each wood-cleated fiberboard box shall not exceed 200 pounds. The gross weight and the size of the fiberboard boxes shall be governed by the applicable box specification. The gross weight of each container shall not exceed 200 pounds. Wood-cleated fiberboard boxes shall be strapped in accordance with the appendix to the box specification. Each fiberboard box shall be reinforced with bands of tape in accordance with the appendix of the applicable box specification.

* 5.2.3. Level C. Packages which require overpacking for acceptance by the carrier shall be packed in shipping containers in a manner that will insure safe delivery at the lowest transportation and packaging charges. The containers shall conform to Uniform Freight Classification Rules and Regulations or other rules and regulations applicable to the mode of transportation.

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*5.3 Marking. In addition to any special marking required by the contract or order (see 6.2), marking for shipment shall be in accordance with MIL-STD-129 and shall include the date of manufacturer, month and year, of the life preservers. In addition, regardless of the mode(s) of transportation, the unit, the intermediate, and the shipping containers shall be marked with the name of the dangerous content "NONFLAMMABLE COMPRESSED GAS," and the proper shipping name "CARBON DIOXIDE LIQUIFIED" shall be conspicuously and legibly marked as prescribed in T. C. George's Tariff No. 13, Interstate Commerce Commission Regulation, paragraph 73.302.

6. NOTES

6.1 Intended use. The life preserver covered by this specification is intended to be used as emergency flotation gear by adult passengers aboard aircraft of the Military Airlift Command during flights over water.

*6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification

(b) Selection of inspection activity for sampling plan C (see 4.5.2.4) and first article samples (see 4.6.1.1)

(c) Disposition of samples (see 4.6.1.3 and 6.6)

(d) Request for results of first article inspection (see 4.6.1.4)

(e) Selection of applicable levels of preservation and packaging and packing

(f) Requirements for additional marking (see 5.3).

6.3 Standard color sample. Requests for access to standard color sample should be addressed to the Systems Engineering Group, Attention: SEMCE, Wright-Patterson Air Force Base, Ohio.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products, which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligble to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Systems Engineering Group, Attention: SEMCE, Wright-Patterson Air Force Base, Ohio 45433, and information pertaining to qualification of products may be obtained from that activity.

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6.5 Acceptable sources for signal light. The following manufacturers' part numbers will be considered acceptable for use as the emergency signal light (see 3.7.4). Use of a signal light other than those listed herein shall require the approval of the procuring activity.

(a) Part No. L-8-1 (ACR Electronics Corp., 551 W. 22nd St., New York 11, New York)

(b) Part No. L-8 (McMurdo Instruments Co., Limited, Ashtead, Surrey, England)

(c) Part No. EW-104-WL-12-12 (Eureka-Williams Corp., Bloomington, Illinois).

6.6 Disposition of samples. The disposition of the samples shall be as follows: One approved life preserver shall be returned to the manufacturer for use in monitoring production. A second approved life preserver shall be returned to the manufacturer along with the final sample as required by sampling plan C (see 4.5.2.4). The other (third) life preserver shall be consumed or destroyed in the first article inspection.

6.7 Color of cord. When cord conforming to MIL-C-5040 cannot be easily obtained in the color specified, white cord may be substituted.

6.8 Identification of changes. The outside margins of this specification have been marked to indicate where changes (deletions, additions, etc) from the previous issue have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in those notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 11

Preparing activity:
Air Force - 11

Review activities:
Air Force - 11, 84

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO 15419

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